
Phase 4 – Advanced SQL Engineering

Project: Enterprise ERP Analytics Implementation

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Mandatory SQL Tasks :-

25 Complex JOIN Queries :-

1.

```
SELECT a.order_id, a.company_name, b.product_name
FROM logistics_erp a
JOIN logistics_erp b ON a.order_id = b.order_id;
```

```
mysql> SELECT a.order_id, a.company_name, b.product_name
-> FROM logistics_erp a
-> JOIN logistics_erp b ON a.order_id = b.order_id;
```

order_id	company_name	product_name
ORD-26-874468	Liberty Industries Inc.	Hydraulic Pump
ORD-26-225204	Nova Industries Inc.	Fuel Pump
ORD-26-810513	Liberty Industries Inc.	Radiator
ORD-26-426696	Summit Industries Inc.	Plastic Pallet
ORD-26-397420	Summit Industries Inc.	Ball Bearing
ORD-26-458515	Apex Industries Inc.	Stretch Film
ORD-26-853878	Apex Industries Inc.	Stretch Film
ORD-26-464469	Liberty Industries Inc.	Bubble Wrap
ORD-26-995570	Nova Industries Inc.	Hydraulic Pump
ORD-26-850685	Horizon Industries Inc.	Face Mask
ORD-26-327353	Apex Industries Inc.	Syringe
ORD-26-984752	Horizon Industries Inc.	Hydraulic Pump
ORD-26-744898	Prime Industries Inc.	Fuel Pump
ORD-26-392420	Summit Industries Inc.	Thermometer
ORD-26-974229	Vertex Industries Inc.	Circuit Breaker
ORD-26-263965	Summit Industries Inc.	Oil Filter
ORD-26-839529	Pioneer Industries Inc.	Ball Bearing
ORD-26-310237	Vertex Industries Inc.	Packing Tape
ORD-26-512320	Pioneer Industries Inc.	Syringe
ORD-26-693706	Prime Industries Inc.	IV Set
ORD-26-519591	Horizon Industries Inc.	Hydraulic Pump
ORD-26-195161	Atlas Industries Inc.	Syringe
ORD-26-987042	Horizon Industries Inc.	Bubble Wrap
ORD-26-163196	Prime Industries Inc.	Surgical Gloves
ORD-26-606489	Prime Industries Inc.	Packing Tape
ORD-26-534691	Apex Industries Inc.	Packing Tape
ORD-26-618023	Vertex Industries Inc.	Surgical Gloves
ORD-26-797154	Nova Industries Inc.	Ethernet Switch
ORD-26-898046	Nova Industries Inc.	Thermometer
ORD-26-539304	Horizon Industries Inc.	Brake Disc
ORD-26-803949	Pioneer Industries Inc.	Servo Motor
ORD-26-716559	Prime Industries Inc.	Corrugated Box
ORD-26-379501	Vertex Industries Inc.	Corrugated Box
ORD-26-331480	Apex Industries Inc.	Ethernet Switch
ORD-26-657845	Liberty Industries Inc.	Valve Assembly
ORD-26-760571	Liberty Industries Inc.	Servo Motor
ORD-26-268277	Prime Industries Inc.	Circuit Breaker
ORD-26-146466	Prime Industries Inc.	Air Compressor
ORD-26-538060	Prime Industries Inc.	Valve Assembly

2. SELECT a.customer_id, COUNT(DISTINCT b.product_id) total_products
FROM logistics_erp a

JOIN logistics_erp b ON a.customer_id = b.customer_id

GROUP BY a.customer_id;

```
mysql> SELECT a.customer_id, COUNT(DISTINCT b.product_id)
-> FROM logistics_erp a
-> JOIN logistics_erp b ON a.customer_id = b.customer_id
-> GROUP BY a.customer_id;
```

customer_id	COUNT(DISTINCT b.product_id)
CUST000066	358
CUST000206	356
CUST000235	380
CUST000487	315
CUST000524	326
CUST000546	342
CUST000566	369
CUST000626	368
CUST000752	350
CUST000785	336

10 rows in set (3.06 sec)

```
mysql>
```

```
3 SELECT a.product_category, SUM(b.order_value) total_sales
FROM logistics_erp a
JOIN logistics_erp b ON a.product_id = b.product_id
GROUP BY a.product_category;
```

```
mysql> SELECT a.product_category, SUM(b.order_value)
-> FROM logistics_erp a
-> JOIN logistics_erp b ON a.product_id = b.product_id
-> GROUP BY a.product_category;
```

product_category	SUM(b.order_value)
Mechanical	77651024.16
Automotive	87064531.08
Packaging	85802722.20
Medical	85818709.34
Electrical	82342546.30

```
5 rows in set (0.02 sec)
```

```
4. SELECT a.courier, COUNT(b.order_id)
FROM logistics_erp a
JOIN logistics_erp b ON a.courier = b.courier
GROUP BY a.courier;
```

```
mysql> SELECT a.courier, COUNT(b.order_id)
-> FROM logistics_erp a
-> JOIN logistics_erp b ON a.courier = b.courier
-> GROUP BY a.courier;
```

courier	COUNT(b.order_id)
Old Dominion	203401
R+L Carriers	168100
Estes Express	215296
FedEx	225625
UPS	164836
XPO Logistics	180625
DHL Express	198025
USPS	179776

```
8 rows in set (0.83 sec)
```

```
5. SELECT a.warehouse, SUM(b.closing_qty)
FROM logistics_erp a
JOIN logistics_erp b ON a.warehouse = b.warehouse
GROUP BY a.warehouse;
```

```
mysql> SELECT a.warehouse, SUM(b.closing_qty)
-> FROM logistics_erp a
-> JOIN logistics_erp b ON a.warehouse = b.warehouse
-> GROUP BY a.warehouse;
```

warehouse	SUM(b.closing_qty)
Dallas DC	348354630
Seattle DC	420432080
Chicago DC	398244680
Atlanta DC	425974154
Phoenix DC	355988734

```
5 rows in set (1.59 sec)
```

```
6. SELECT a.order_id, b.dispatch_date, b.delivered_date  
FROM logistics_erp a  
JOIN logistics_erp b ON a.order_id = b.order_id;
```

```
mysql> SELECT a.order_id, b.dispatch_date, b.delivered_date
-> FROM logistics_erp a
-> JOIN logistics_erp b ON a.order_id = b.order_id;
```

order_id	dispatch_date	delivered_date
ORD-26-100771	2026-06-24	NULL
ORD-26-101240	2026-01-29	2026-01-31
ORD-26-101244	2026-10-04	NULL
ORD-26-101372	2026-06-17	2026-06-22
ORD-26-101468	2026-11-02	NULL
ORD-26-102190	2026-01-15	NULL
ORD-26-102317	2026-04-28	NULL
ORD-26-102351	2026-05-30	2026-01-06
ORD-26-102654	2026-06-16	NULL
ORD-26-103091	2026-07-06	NULL
ORD-26-103093	2026-04-21	2026-04-23
ORD-26-103377	2026-05-24	2026-05-25
ORD-26-103463	2026-01-24	2026-01-28
ORD-26-103563	2026-02-27	NULL
ORD-26-103741	2026-07-05	2026-11-05
ORD-26-103948	2026-04-20	2026-04-25
ORD-26-104236	2026-03-01	NULL
ORD-26-104654	2026-03-20	2026-03-23
ORD-26-105049	2026-06-22	2026-06-24
ORD-26-105362	2026-01-14	2026-01-15
ORD-26-105547	2026-06-04	2026-10-04
ORD-26-105663	2026-05-29	2026-02-06
ORD-26-106049	2026-05-22	2026-05-28
ORD-26-106299	2026-03-27	NULL
ORD-26-107297	2026-06-25	2026-06-28
ORD-26-107474	2026-05-03	2026-06-03
ORD-26-107507	2026-01-04	NULL
ORD-26-107633	2026-03-01	2026-06-01
ORD-26-107650	2026-04-06	2026-06-06
ORD-26-107893	2026-02-03	NULL
ORD-26-107900	2026-03-03	NULL
ORD-26-108528	2026-02-24	NULL
ORD-26-108892	2026-04-27	2026-04-29
ORD-26-108935	2026-04-29	NULL

7. SELECT a.customer_id, AVG(b.order_value)

FROM logistics_erp a

JOIN logistics_erp b ON a.customer_id = b.customer_id

GROUP BY a.customer_id;

```
mysql> SELECT a.customer_id, AVG(b.order_value)
-> FROM logistics_erp a
-> JOIN logistics_erp b ON a.customer_id = b.customer_id
-> GROUP BY a.customer_id;
```

customer_id	AVG(b.order_value)
CUST000546	112842.023889
CUST000752	112470.954800
CUST000066	114070.543268
CUST000785	127180.446756
CUST000487	121425.094794
CUST000206	130885.357612
CUST000566	124586.634472
CUST000626	113201.884076
CUST000524	117033.492331
CUST000235	122434.394737

10 rows in set (1.73 sec)

```
8. SELECT a.product_id, MAX(b.order_qty)
FROM logistics_erp a
JOIN logistics_erp b ON a.product_id = b.product_id
GROUP BY a.product_id;
```

```
mysql> SELECT a.product_id, MAX(b.order_qty)
-> FROM logistics_erp a
-> JOIN logistics_erp b ON a.product_id = b.product_id
-> GROUP BY a.product_id;
```

product_id	MAX(b.order_qty)
PRD-003IZB3MCR	51
PRD-013GAF4IU1	136
PRD-019SY5F4B8	205
PRD-01WEVBW8EN	99
PRD-025NU81D1A	325
PRD-02Q4J0YQ8R	396
PRD-03D65YT8T0	63
PRD-03DU22NKPU	447
PRD-03KSPFVRF4	479
PRD-03LHT5TDTZ	438
PRD-03ZA5SNVA4	281
PRD-054XDI5YJA	102
PRD-05D97KHIS0	435
PRD-05IQWFOGFH	457
PRD-05JD815T72	162
PRD-05QGSC0H6R	72
PRD-065G4PPB1C	22
PRD-066RY71JPO	132
PRD-06B62PW06Q	445
PRD-06MY4JI5X8	327
PRD-07QWG4KIMR	343
PRD-07Y48UAWVL	106
PRD-08HNAX5PM8	131
PRD-08RZ3BQ9HX	340

09. SELECT a.customer_id, COUNT(b.order_id)

FROM logistics_erp a

JOIN logistics_erp b ON a.customer_id = b.customer_id

GROUP BY a.customer_id;

```
mysql> SELECT a.customer_id, COUNT(b.order_id)
-> FROM logistics_erp a
-> JOIN logistics_erp b ON a.customer_id = b.customer_id
-> GROUP BY a.customer_id;
```

customer_id	COUNT(b.order_id)
CUST000546	116964
CUST000752	122500
CUST000066	128164
CUST000785	112896
CUST000487	99225
CUST000206	126736
CUST000566	136161
CUST000626	135424
CUST000524	106276
CUST000235	144400

10 rows in set (1.62 sec)

10. SELECT a.product_category, COUNT(DISTINCT b.customer_id)
FROM logistics_erp a

JOIN logistics_erp b ON a.product_category = b.product_category

GROUP BY a.product_category;

```
mysql> SELECT a.product_category, COUNT(DISTINCT b.customer_id)
-> FROM logistics_erp a
-> JOIN logistics_erp b ON a.product_category = b.product_category
-> GROUP BY a.product_category;
```

product_category	COUNT(DISTINCT b.customer_id)
Automotive	10
Electrical	10
Mechanical	10
Medical	10
Packaging	10

5 rows in set (3.24 sec)

```
11. SELECT a.order_date, SUM(b.order_value)
FROM logistics_erp a
JOIN logistics_erp b ON a.order_date = b.order_date
GROUP BY a.order_date;
```

```
mysql> SELECT a.order_date, SUM(b.order_value)
-> FROM logistics_erp a
-> JOIN logistics_erp b ON a.order_date = b.order_date
-> GROUP BY a.order_date;
```

order_date	SUM(b.order_value)
2026-06-28	23474614.72
2026-08-04	22500866.64
2026-04-21	48366782.80
2026-04-30	22564485.58
2026-11-02	17847561.12
2026-11-06	33333212.64
2026-03-01	29754950.68
2026-10-01	27614626.02
2026-05-31	72813969.58
2026-05-02	47358428.46
2026-03-14	33001346.08
2026-04-29	43329636.29
2026-12-01	66963614.13
2026-03-02	43077067.04
2026-05-14	68445779.75
2026-01-19	53632617.62
2026-05-15	30866558.64

```
12 . SELECT a.shipment_status, COUNT(b.order_id)
FROM logistics_erp a
JOIN logistics_erp b ON a.shipment_status = b.shipment_status
GROUP BY a.shipment_status;
```

```
mysql> SELECT a.shipment_status, COUNT(b.order_id)
-> FROM logistics_erp a
-> JOIN logistics_erp b ON a.shipment_status = b.shipment_status
-> GROUP BY a.shipment_status;
```

shipment_status	COUNT(b.order_id)
In Transit	240100
Pending Dispatch	249001
Delivered	2232036
Delayed	225625
Out for Delivery	293764

```
5 rows in set (1.72 sec)
```

```
13. SELECT a.product_name, SUM(b.order_qty)
FROM logistics_erp a
JOIN logistics_erp b ON a.product_name = b.product_name
GROUP BY a.product_name;
```



```
mysql> SELECT a.product_name, SUM(b.order_qty)
-> FROM logistics_erp a
-> JOIN logistics_erp b ON a.product_name = b.product_name
-> GROUP BY a.product_name;
```

product_name	SUM(b.order_qty)
Hydraulic Pump	3934080
Fuel Pump	3868160
Radiator	4582380
Plastic Pallet	4368592
Ball Bearing	5964658
Stretch Film	5830350
Bubble Wrap	6491200
Face Mask	4335276
Syringe	4330860
Thermometer	5857362
Circuit Breaker	5009902
Oil Filter	5833946
Packing Tape	4831408
IV Set	4951240
Surgical Gloves	5252668
Ethernet Switch	4677002
Brake Disc	5475260
Servo Motor	5230113
Corrugated Box	3846528
Valve Assembly	4850266
Air Compressor	4501192
PLC Controller	5002281
Shock Absorber	5399040
Industrial Control Panel	5176171
Gearbox	3678885

25 rows in set (0.43 sec)

14 SELECT a.customer_id, MIN(b.order_value)

FROM logistics_erp a

JOIN logistics_erp b ON a.customer_id = b.customer_id

GROUP BY a.customer_id;

```
mysql> SELECT a.customer_id, MIN(b.order_value)
-> FROM logistics_erp a
-> JOIN logistics_erp b ON a.customer_id = b.customer_id
-> GROUP BY a.customer_id;
```

customer_id	MIN(b.order_value)
CUST000546	896.40
CUST000752	658.08
CUST000066	236.65
CUST000785	733.59
CUST000487	822.15
CUST000206	393.99
CUST000566	73.04
CUST000626	359.75
CUST000524	299.04
CUST000235	1127.97

10 rows in set (2.29 sec)

```
15. SELECT a.product_id, AVG(b.order_qty)
FROM logistics_erp a
JOIN logistics_erp b ON a.product_id = b.product_id
GROUP BY a.product_id;
```

```
mysql> SELECT a.product_id, AVG(b.order_qty)
-> FROM logistics_erp a
-> JOIN logistics_erp b ON a.product_id = b.product_id
-> GROUP BY a.product_id;
```

product_id	AVG(b.order_qty)
PRD-003IZB3MCR	51.0000
PRD-013GAF4IU1	136.0000
PRD-019SY5F4B8	205.0000
PRD-01WEVBW8EN	99.0000
PRD-025NU81D1A	325.0000
PRD-02Q4J0YQ8R	396.0000
PRD-03D65YT8T0	63.0000
PRD-03DU22NKPU	447.0000
PRD-03KSPFVRF4	479.0000
PRD-03LHT5TDTZ	438.0000
PRD-03ZA5SNVA4	281.0000
PRD-054XDI5YJA	102.0000
PRD-05D97KHISO	435.0000
PRD-05IQWFOGFH	457.0000
PRD-05JD815T72	162.0000
PRD-05QGSCOH6R	72.0000
PRD-065G4PPB1C	22.0000
PRD-066RY71JPO	132.0000
PRD-06B62PW06Q	445.0000
PRD-06MY4JI5X8	327.0000
PRD-07QWG4KIMR	343.0000
PRD-07Y48UAWVL	106.0000
PRD-08HNAX5PM8	131.0000
PRD-08RZ3BQ9HX	340.0000
PRD-08YL3TZACU	212.0000
PRD-092NHUMTMF	247.0000
PRD-09H2RXMX2G	178.0000
PRD-09HA19SQHV	131.0000
PRD-09HX7YBUIC	174.0000
PRD-09UPZ4EDBZ	343.0000
PRD-09W5ENISF6	125.0000

16. SELECT a.courier, AVG(DATEDIFF(b.delivered_date, b.dispatch_date))
FROM logistics_erp a

JOIN logistics_erp b ON a.courier = b.courier

GROUP BY a.courier;

```
mysql> SELECT a.courier, AVG(DATEDIFF(b.delivered_date, b.dispatch_date))  
-> FROM logistics_erp a  
-> JOIN logistics_erp b ON a.courier = b.courier  
-> GROUP BY a.courier;
```

courier	AVG(DATEDIFF(b.delivered_date, b.dispatch_date))
Old Dominion	1.2204
R+L Carriers	8.5533
Estes Express	4.0000
FedEx	2.4091
UPS	-1.9673
XPO Logistics	1.5777
DHL Express	-3.9300
USPS	3.4268

8 rows in set (1.05 sec)

17. SELECT a.customer_id, COUNT(DISTINCT b.order_date)

FROM logistics_erp a JOIN logistics_erp b ON a.customer_id =
b.customer_id

GROUP BY a.customer_id;

```
mysql> SELECT a.customer_id, COUNT(DISTINCT b.order_date)
-> FROM logistics_erp a
-> JOIN logistics_erp b ON a.customer_id = b.customer_id
-> GROUP BY a.customer_id;
```

customer_id	COUNT(DISTINCT b.order_date)
CUST000066	151
CUST000206	160
CUST000235	153
CUST000487	149
CUST000524	155
CUST000546	159
CUST000566	159
CUST000626	162
CUST000752	162
CUST000785	149

10 rows in set (2.21 sec)

18. SELECT a.product_category, MAX(b.order_value)

FROM logistics_erp a

JOIN logistics_erp b ON a.product_category = b.product_category

GROUP BY a.product_category;

```
mysql>
mysql> SELECT a.product_category, MAX(b.order_value)
-> FROM logistics_erp a
-> JOIN logistics_erp b ON a.product_category = b.product_category
-> GROUP BY a.product_category;
```

product_category	MAX(b.order_value)
Mechanical	453816.45
Automotive	464297.54
Packaging	460521.16
Medical	465515.64
Electrical	457922.32

5 rows in set (1.48 sec)

19. SELECT a.warehouse, COUNT(DISTINCT b.product_id)

FROM logistics_erp a

JOIN logistics_erp b ON a.warehouse = b.warehouse

GROUP BY a.warehouse;

```
mysql>
mysql> SELECT a.warehouse, COUNT(DISTINCT b.product_id)
-> FROM logistics_erp a
-> JOIN logistics_erp b ON a.warehouse = b.warehouse
-> GROUP BY a.warehouse;
```

warehouse	COUNT(DISTINCT b.product_id)
Atlanta DC	733
Chicago DC	710
Dallas DC	666
Phoenix DC	673
Seattle DC	718

5 rows in set (37.03 sec)

20. SELECT a.customer_id, SUM(b.order_qty)

FROM logistics_erp a

JOIN logistics_erp b ON a.customer_id = b.customer_id

GROUP BY a.customer_id;

```
mysql>
mysql> SELECT a.customer_id, SUM(b.order_qty)
      -> FROM logistics_erp a
      -> JOIN logistics_erp b ON a.customer_id = b.customer_id
      -> GROUP BY a.customer_id;
```

customer_id	SUM(b.order_qty)
CUST000546	29201328
CUST000752	30188200
CUST000066	30910436
CUST000785	27928656
CUST000487	25250715
CUST000206	32247904
CUST000566	34576038
CUST000626	33190656
CUST000524	27203722
CUST000235	36875200

10 rows in set (1.80 sec)

21 SELECT a.product_id, SUM(b.order_value)

FROM logistics_erp a

JOIN logistics_erp b ON a.product_id = b.product_id

GROUP BY a.product_id;

```
mysql> SELECT a.product_id, SUM(b.order_value)
-> FROM logistics_erp a
-> JOIN logistics_erp b ON a.product_id = b.product_id
-> GROUP BY a.product_id;
```

product_id	SUM(b.order_value)
PRD-003IZB3MCR	29435.67
PRD-013GAF4IU1	89656.64
PRD-019SY5F4B8	81413.70
PRD-01WEVBW8EN	58377.33
PRD-025NU81D1A	59475.00
PRD-02Q4J0YQ8R	88094.16
PRD-03D65YT8T0	53219.88
PRD-03DU22NKPU	155444.25
PRD-03KSPFVRF4	395529.46
PRD-03LHT5TDTZ	226945.32
PRD-03ZA5SNVA4	211497.46
PRD-054XDI5YJA	42252.48
PRD-05D97KHISO	388159.20
PRD-05IQWFOGFH	248018.47
PRD-05JD815T72	99709.38
PRD-05QGSCOH6R	46388.88
PRD-065G4PPB1C	8925.18
PRD-066RY71JPO	44502.48
PRD-06B62PW06Q	206706.95
PRD-06MY4JI5X8	264870.00
PRD-07QWG4KIMR	137896.29
PRD-07Y48UAWVL	87660.94
PRD-08HNA55PM8	124152.62

```

22 SELECT a.order_id, COUNT(b.product_id)
FROM logistics_erp a
JOIN logistics_erp b ON a.order_id = b.order_id
GROUP BY a.order_id;

```

```
mysql> SELECT a.order_id, COUNT(b.product_id)
-> FROM logistics_erp a
-> JOIN logistics_erp b ON a.order_id = b.order_id
-> GROUP BY a.order_id;
```

order_id	COUNT(b.product_id)
ORD-26-100771	1
ORD-26-101240	1
ORD-26-101244	1
ORD-26-101372	1
ORD-26-101468	1
ORD-26-102190	1
ORD-26-102317	1
ORD-26-102351	1
ORD-26-102654	1
ORD-26-103091	1
ORD-26-103093	1
ORD-26-103377	1
ORD-26-103463	1
ORD-26-103563	1
ORD-26-103741	1

```

23 SELECT a.product_category, AVG(b.order_value)
FROM logistics_erp a
JOIN logistics_erp b ON a.product_category = b.product_category
GROUP BY a.product_category;

```

```
mysql> SELECT a.product_category, AVG(b.order_value)
-> FROM logistics_erp a
-> JOIN logistics_erp b ON a.product_category = b.product_category
-> GROUP BY a.product_category;
```

product_category	AVG(b.order_value)
Mechanical	114192.682588
Automotive	122799.056530
Packaging	119836.204190
Medical	122949.440315
Electrical	118138.516930

5 rows in set (1.79 sec)

24 SELECT a.customer_id, MAX(b.order_date)

FROM logistics_erp a

JOIN logistics_erp b ON a.customer_id = b.customer_id

GROUP BY a.customer_id;

```
mysql> SELECT a.customer_id, MAX(b.order_date)
-> FROM logistics_erp a
-> JOIN logistics_erp b ON a.customer_id = b.customer_id
-> GROUP BY a.customer_id;
```

customer_id	MAX(b.order_date)
CUST000546	2026-12-06
CUST000752	2026-12-06
CUST000066	2026-12-06
CUST000785	2026-12-06
CUST000487	2026-12-06
CUST000206	2026-12-06
CUST000566	2026-12-06
CUST000626	2026-12-06
CUST000524	2026-12-06
CUST000235	2026-12-06

10 rows in set (1.65 sec)

```
25 SELECT a.courier, SUM(b.order_value)
FROM logistics_erp a
JOIN logistics_erp b ON a.courier = b.courier
GROUP BY a.courier;
```

```
mysql> SELECT a.courier, SUM(b.order_value)
-> FROM logistics_erp a
-> JOIN logistics_erp b ON a.courier = b.courier
-> GROUP BY a.courier;
```

courier	SUM(b.order_value)
Old Dominion	24031320066.93
R+L Carriers	20008757831.70
Estes Express	25047717265.92
FedEx	27680392136.00
UPS	20365186345.00
XPO Logistics	22110765411.50
DHL Express	22479892486.85
USPS	21892816860.72

```
8 rows in set (1.01 sec)
```

15 Common Table Expressions (CTEs)

1. WITH cte1 AS (SELECT customer_id, SUM(order_value) total_sales
FROM logistics_erp

GROUP BY customer_id)

SELECT * FROM cte1;

```
mysql> WITH cte1 AS (
  -> SELECT customer_id, SUM(order_value) total_sales
  -> FROM logistics_erp GROUP BY customer_id)
  -> SELECT * FROM cte1;
```

customer_id	total_sales
CUST000066	40837254.49
CUST000206	46595187.31
CUST000235	46525070.00
CUST000487	38248904.86
CUST000524	38152918.50
CUST000546	38591972.17
CUST000566	45972468.12
CUST000626	41658293.34
CUST000752	39364834.18
CUST000785	42732630.11

```
10 rows in set (0.01 sec)
```

2. WITH cte2 AS

```
( SELECT product_id, SUM(order_qty) qty
FROM logistics_erp
GROUP BY product_id)
SELECT * FROM cte2;
```

```
mysql> WITH cte2 AS (
  -> SELECT product_id, SUM(order_qty) qty
  -> FROM logistics_erp GROUP BY product_id)
  -> SELECT * FROM cte2;
```

product_id	qty
PRD-003IZB3MCR	51
PRD-013GAF4IU1	136
PRD-019SY5F4B8	205
PRD-01WEVBW8EN	99
PRD-025NU81D1A	325
PRD-02Q4J0YQ8R	396
PRD-03D65YT8T0	63
PRD-03DU22NKPU	447
PRD-03KSPFVRF4	479
PRD-03LHT5TDTZ	438
PRD-03ZA5SNVA4	281
PRD-054XDI5YJA	102
PRD-05D97KHISO	435
PRD-05IQWFOGFH	457
PRD-05JD815T72	162
PRD-05QGSC0H6R	72
PRD-065G4PPB1C	22
PRD-066RY71JPO	132
PRD-06B62PW06Q	445
PRD-06MY4JI5X8	327
PRD-07QWG4KIMR	343
PRD-07Y48UAWVL	106
PRD-08HNAX5PM8	131
PRD-08RZ3BQ9HX	340
PRD-08YL3TZACU	212

3. WITH cte3 AS

```
( SELECT courier, COUNT(*) cnt
```

```
FROM logistics_erp
```

```
GROUP BY courier)
```

```
SELECT * FROM cte3;
```

```
mysql> WITH cte3 AS (
  -> SELECT courier, COUNT(*) cnt
  -> FROM logistics_erp GROUP BY courier)
  -> SELECT * FROM cte3;
```

courier	cnt
Old Dominion	451
R+L Carriers	410
Estes Express	464
FedEx	475
UPS	406
XPO Logistics	425
DHL Express	445
USPS	424

8 rows in set (0.01 sec)

4. WITH cte4 AS

(SELECT warehouse, SUM(closing_qty) stock

FROM logistics_erp GROUP BY warehouse) SELECT * FROM cte4;

```
mysql> WITH cte4 AS (
  -> SELECT warehouse, SUM(closing_qty) stock
  -> FROM logistics_erp GROUP BY warehouse)
  -> SELECT * FROM cte4;
```

warehouse	stock
Dallas DC	523055
Seattle DC	585560
Chicago DC	560908
Atlanta DC	581138
Phoenix DC	528958

5 rows in set (0.01 sec)

5. WITH cte5 AS

```
( SELECT product_category, AVG(order_value) avg_val  
FROM logistics_erp  
GROUP BY product_category)  
  
SELECT * FROM cte5;
```

```
mysql> WITH cte5 AS (  
-> SELECT product_category, AVG(order_value) avg_val  
-> FROM logistics_erp GROUP BY product_category)  
-> SELECT * FROM cte5;  
+-----+-----+  
| product_category | avg_val |  
+-----+-----+  
| Mechanical      | 114192.682588 |  
| Automotive      | 122799.056530 |  
| Packaging        | 119836.204190 |  
| Medical         | 122949.440315 |  
| Electrical      | 118138.516930 |  
+-----+-----+  
5 rows in set (0.01 sec)
```

6. WITH cte6 AS

```
( SELECT order_id, DATEDIFF(delivered_date, dispatch_date) delay  
FROM logistics_erp)  
  
SELECT * FROM cte6;
```

```
mysql> WITH cte6 AS (
  -> SELECT order_id, DATEDIFF(delivered_date, dispatch_date) delay
  -> FROM logistics_erp)
  -> SELECT * FROM cte6;
```

order_id	delay
ORD-26-874468	NULL
ORD-26-225204	NULL
ORD-26-810513	1
ORD-26-426696	-56
ORD-26-397420	NULL
ORD-26-458515	NULL
ORD-26-853878	NULL
ORD-26-464469	NULL
ORD-26-995570	90
ORD-26-850685	92
ORD-26-327353	5
ORD-26-984752	59
ORD-26-744898	NULL
ORD-26-392420	31
ORD-26-974229	NULL
ORD-26-263965	NULL
ORD-26-839529	1
ORD-26-310237	-290
ORD-26-512320	NULL
ORD-26-693706	NULL
ORD-26-519591	NULL
ORD-26-195161	3
ORD-26-987042	NULL
ORD-26-163196	NULL
ORD-26-606489	NULL
ORD-26-534691	4
ORD-26-618023	-201
ORD-26-707154	NULL

7. WITH cte7 AS

```
( SELECT customer_id, COUNT(order_id) cnt
```

```
FROM logistics_erp
```

```
GROUP BY customer_id)
```

```
SELECT * FROM cte7;
```

```
mysql> WITH cte7 AS (  
-> SELECT customer_id, COUNT(order_id) cnt  
-> FROM logistics_erp GROUP BY customer_id)  
-> SELECT * FROM cte7;
```

customer_id	cnt
CUST000066	358
CUST000206	356
CUST000235	380
CUST000487	315
CUST000524	326
CUST000546	342
CUST000566	369
CUST000626	368
CUST000752	350
CUST000785	336

10 rows in set (0.01 sec)

8. WITH cte8 AS

```
( SELECT product_name, SUM(order_qty) qty  
FROM logistics_erp  
GROUP BY product_name)  
SELECT * FROM cte8;
```

```
mysql> WITH cte8 AS (
  -> SELECT product_name, SUM(order_qty) qty
  -> FROM logistics_erp GROUP BY product_name)
  -> SELECT * FROM cte8;
```

product_name	qty
Hydraulic Pump	30735
Fuel Pump	30220
Radiator	34715
Plastic Pallet	32122
Ball Bearing	37751
Stretch Film	38869
Bubble Wrap	40570
Face Mask	32843
Syringe	33060
Thermometer	39846
Circuit Breaker	35281
Oil Filter	39154
Packing Tape	34024
IV Set	35366
Surgical Gloves	35491
Ethernet Switch	34903
Brake Disc	36995
Servo Motor	37093
Corrugated Box	30051
Valve Assembly	34894
Air Compressor	33097
PLC Controller	36513
Shock Absorber	35520
Industrial Control Panel	36197
Gearbox	30915

25 rows in set (0.01 sec)

9. WITH cte9 AS

(SELECT order_date, SUM(order_value) sales

FROM logistics_erp

GROUP BY order_date)

SELECT * FROM cte9;

```
mysql> WITH cte9 AS (  
-> SELECT order_date, SUM(order_value) sales  
-> FROM logistics_erp GROUP BY order_date)  
-> SELECT * FROM cte9;
```

order_date	sales
2026-06-28	1467163.42
2026-08-04	1607204.76
2026-04-21	2418339.14
2026-04-30	1611748.97
2026-11-02	1115472.57
2026-11-06	2083325.79
2026-03-01	2125353.62
2026-10-01	1534145.89
2026-05-31	3309725.89
2026-05-02	2255163.26
2026-03-14	2062584.13
2026-04-29	1883897.23
2026-12-01	3188743.53
2026-03-02	2533945.12
2026-05-14	2737831.19
2026-01-19	2331852.94
2026-05-15	1815679.92
2026-08-01	1530393.94
2026-03-06	2345953.68
2026-04-23	1774273.83
2026-01-21	1729098.65
2026-06-24	2389312.93
2026-05-21	1242717.31
2026-04-22	2175915.49
2026-02-21	2349651.31
2026-01-26	3324620.44
2026-09-05	3094557.63
2026-05-25	1012428.26
2026-05-05	1227602.26
2026-02-28	1952655.53
2026-02-03	1810316.16
2026-04-01	2176945.22
2026-10-04	2953129.33
2026-07-01	1835822.64

10. WITH cte10 AS

(SELECT shipment_status, COUNT(*) cnt

FROM logistics_erp

GROUP BY shipment_status)

SELECT * FROM cte10;

```
mysql> WITH cte10 AS (  
-> SELECT shipment_status, COUNT(*) cnt  
-> FROM logistics_erp GROUP BY shipment_status)  
-> SELECT * FROM cte10;  
+-----+-----+  
| shipment_status | cnt |  
+-----+-----+  
| In Transit      | 490 |  
| Pending Dispatch | 499 |  
| Delivered       | 1494 |  
| Delayed         | 475 |  
| Out for Delivery | 542 |  
+-----+-----+  
5 rows in set (0.01 sec)
```

11. WITH cte11 AS

(SELECT customer_id, MAX(order_value) max_val

FROM logistics_erp

GROUP BY customer_id)

SELECT * FROM cte11;

```
mysql> WITH cte11 AS (
  -> SELECT customer_id, MAX(order_value) max_val
  -> FROM logistics_erp GROUP BY customer_id)
  -> SELECT * FROM cte11;
```

customer_id	max_val
CUST000066	465515.64
CUST000206	438294.24
CUST000235	440289.90
CUST000487	457922.32
CUST000524	397443.51
CUST000546	460521.16
CUST000566	464297.54
CUST000626	440822.40
CUST000752	453816.45
CUST000785	454752.00

```
10 rows in set (0.01 sec)
```

12. WITH cte12 AS

```
( SELECT product_id, MIN(order_qty) min_qty
FROM logistics_erp
GROUP BY product_id)
SELECT * FROM cte12;
```

```
mysql> WITH cte12 AS (
  -> SELECT product_id, MIN(order_qty) min_qty
  -> FROM logistics_erp GROUP BY product_id)
  -> SELECT * FROM cte12;
```

product_id	min_qty
PRD-003IZB3MCR	51
PRD-013GAF4IU1	136
PRD-019SY5F4B8	205
PRD-01WEVBW8EN	99
PRD-025NU81D1A	325
PRD-02Q4J0YQ8R	396
PRD-03D65YT8T0	63
PRD-03DU22NKPU	447
PRD-03KSPFVRF4	479
PRD-03LHT5TDTZ	438
PRD-03ZA5SNVA4	281
PRD-054XDI5YJA	102
PRD-05D97KHISO	435
PRD-05IQWFOGFH	457
PRD-05JD815T72	162
PRD-05QGSCOH6R	72
PRD-065G4PPB1C	22
PRD-066RY71JPO	132

13.WITH cte13 AS

```
( SELECT warehouse, AVG(closing_qty) avg_stock
```

```
FROM logistics_erp
```

```
GROUP BY warehouse)
```

```
SELECT * FROM cte13;
```



```
mysql> WITH cte13 AS (
  -> SELECT warehouse, AVG(closing_qty) avg_stock
  -> FROM logistics_erp GROUP BY warehouse)
  -> SELECT * FROM cte13;
```

warehouse	avg_stock
Dallas DC	785.3679
Seattle DC	815.5432
Chicago DC	790.0113
Atlanta DC	792.8213
Phoenix DC	785.9703

5 rows in set (0.01 sec)

14. WITH cte14 AS

(SELECT courier, SUM(order_value) total

FROM logistics_erp

GROUP BY courier)

SELECT * FROM cte14;

```
mysql> WITH cte14 AS (
-> SELECT courier, SUM(order_value) total
-> FROM logistics_erp GROUP BY courier)
-> SELECT * FROM cte14;
```

courier	total
Old Dominion	53284523.43
R+L Carriers	48801848.37
Estes Express	53982149.28
FedEx	58274509.76
UPS	50160557.50
XPO Logistics	52025330.38
DHL Express	50516612.33
USPS	51634002.03

```
8 rows in set (0.01 sec)
```

15. WITH cte15 AS

```
( SELECT product_category, COUNT(*) cnt
FROM logistics_erp
GROUP BY product_category)
SELECT * FROM cte15;
```

```
mysql> WITH cte15 AS (  
-> SELECT product_category, COUNT(*) cnt  
-> FROM logistics_erp GROUP BY product_category)  
-> SELECT * FROM cte15;
```

product_category	cnt
Mechanical	680
Automotive	709
Packaging	716
Medical	698
Electrical	697

5 rows in set (0.01 sec)

15 Window Function Queries

--1

```
SELECT order_id, order_value,  
RANK() OVER (ORDER BY order_value DESC) rnk  
FROM logistics_erp;
```

```
mysql> SELECT order_id, order_value,
-> RANK() OVER (ORDER BY order_value DESC) rnk
-> FROM logistics_erp;
```

order_id	order_value	rnk
ORD-26-988208	465515.64	1
ORD-26-499279	464297.54	2
ORD-26-821910	460521.16	3
ORD-26-789254	457922.32	4
ORD-26-410770	455783.30	5
ORD-26-990703	454752.00	6
ORD-26-920128	453816.45	7
ORD-26-782142	452049.44	8
ORD-26-982201	450572.42	9
ORD-26-955537	446467.32	10
ORD-26-367576	443939.90	11
ORD-26-163159	440822.40	12
ORD-26-962922	440289.90	13
ORD-26-408785	439750.50	14
ORD-26-629673	438294.24	15
ORD-26-813315	438225.90	16
ORD-26-832002	437638.40	17
ORD-26-631959	436403.99	18
ORD-26-437150	433250.71	19
ORD-26-696300	431781.24	20
ORD-26-275002	431375.00	21
ORD-26-134858	429862.68	22
ORD-26-895796	429851.60	23
ORD-26-376477	429229.66	24
ORD-26-954202	428314.25	25
ORD-26-783498	428184.70	26
ORD-26-909463	426967.47	27
ORD-26-863208	424491.68	28
ORD-26-233208	424000.94	29
ORD-26-589506	422726.43	30
ORD-26-458887	418637.88	31

-- 2

SELECT order_id, order_value,

DENSE_RANK() OVER (ORDER BY order_value DESC)

FROM logistics_erp;

```
mysql> SELECT order_id, order_value,  
        -> DENSE_RANK() OVER (ORDER BY order_value DESC)  
        -> FROM logistics_erp;
```

order_id	order_value	DENSE_RANK() OVER (ORDER BY order_value DESC)
ORD-26-988208	465515.64	1
ORD-26-499279	464297.54	2
ORD-26-821910	460521.16	3
ORD-26-789254	457922.32	4
ORD-26-410770	455783.30	5
ORD-26-990703	454752.00	6
ORD-26-920128	453816.45	7
ORD-26-782142	452049.44	8
ORD-26-982201	450572.42	9
ORD-26-955537	446467.32	10
ORD-26-367576	443939.90	11
ORD-26-163159	440822.40	12
ORD-26-962922	440289.90	13
ORD-26-408785	439750.50	14
ORD-26-629673	438294.24	15
ORD-26-813315	438225.90	16
ORD-26-832002	437638.40	17
ORD-26-631959	436403.99	18
ORD-26-437150	433250.71	19
ORD-26-696300	431781.24	20
ORD-26-275002	431375.00	21
ORD-26-134858	429862.68	22
ORD-26-895796	429851.60	23
ORD-26-376477	429229.66	24
ORD-26-954202	428314.25	25
ORD-26-783498	428184.70	26
ORD-26-909463	426967.47	27
ORD-26-863208	424491.68	28
ORD-26-233208	424000.94	29

-- 3

SELECT order_id,

ROW_NUMBER() OVER (ORDER BY order_date)

FROM logistics_erp;

```
mysql> SELECT order_id,  
-> ROW_NUMBER() OVER (ORDER BY order_date)  
-> FROM logistics_erp;
```

order_id	ROW_NUMBER() OVER (ORDER BY order_date)
ORD-26-825324	1
ORD-26-882775	2
ORD-26-104236	3
ORD-26-563863	4
ORD-26-807731	5
ORD-26-459760	6
ORD-26-724449	7
ORD-26-844152	8
ORD-26-987169	9
ORD-26-288897	10
ORD-26-107633	11
ORD-26-109915	12
ORD-26-603472	13
ORD-26-636608	14
ORD-26-656558	15
ORD-26-479352	16
ORD-26-542349	17
ORD-26-831874	18
ORD-26-844441	19
ORD-26-875520	20
ORD-26-530342	21
ORD-26-447457	22
ORD-26-950184	23
ORD-26-987103	24
ORD-26-712681	25

-- 4

SELECT customer_id, order_value,

SUM(order_value) OVER (PARTITION BY customer_id)

FROM logistics_erp;

```
mysql> SELECT customer_id, order_value,  
-> SUM(order_value) OVER (PARTITION BY customer_id)  
-> FROM logistics_erp;
```

customer_id	order_value	SUM(order_value) OVER (PARTITION BY customer_id)
CUST000066	92052.50	40837254.49
CUST000066	11281.40	40837254.49
CUST000066	42028.47	40837254.49
CUST000066	155524.86	40837254.49
CUST000066	145783.29	40837254.49
CUST000066	6887.04	40837254.49
CUST000066	23516.64	40837254.49
CUST000066	12110.72	40837254.49
CUST000066	114653.60	40837254.49
CUST000066	100981.85	40837254.49
CUST000066	137353.09	40837254.49
CUST000066	372863.66	40837254.49
CUST000066	98638.54	40837254.49
CUST000066	309895.44	40837254.49
CUST000066	64583.50	40837254.49
CUST000066	79474.50	40837254.49
CUST000066	78097.60	40837254.49
CUST000066	8495.24	40837254.49
CUST000066	5010.62	40837254.49
CUST000066	43512.66	40837254.49
CUST000066	62711.64	40837254.49
CUST000066	11782.43	40837254.49
CUST000066	3982.00	40837254.49
CUST000066	35512.61	40837254.49
CUST000066	185988.16	40837254.49
CUST000066	171045.54	40837254.49
CUST000066	128585.60	40837254.49
CUST000066	36847.38	40837254.49
CUST000066	32178.30	40837254.49
CUST000066	140794.50	40837254.49
CUST000066	45814.88	40837254.49
CUST000066	49220.28	40837254.49
CUST000066	186879.54	40837254.49
CUST000066	36498.51	40837254.49
CUST000066	207470.88	40837254.49
CUST000066	30039.62	40837254.49

-- 5

```
SELECT product_id, order_qty,  
  
AVG(order_qty) OVER (PARTITION BY product_id)  
  
FROM logistics_erp;
```

```
mysql> SELECT product_id, order_qty,  
-> AVG(order_qty) OVER (PARTITION BY product_id)  
-> FROM logistics_erp;
```

product_id	order_qty	AVG(order_qty) OVER (PARTITION BY product_id)
PRD-003IZB3MCR	51	51.0000
PRD-013GAF4IU1	136	136.0000
PRD-019SY5F4B8	205	205.0000
PRD-01WEVBW8EN	99	99.0000
PRD-025NU81D1A	325	325.0000
PRD-02Q4J0YQ8R	396	396.0000
PRD-03D65YT8T0	63	63.0000
PRD-03DU22NKPU	447	447.0000
PRD-03KSPFVRF4	479	479.0000
PRD-03LHT5TDTZ	438	438.0000
PRD-03ZA5SNVA4	281	281.0000
PRD-054XDI5YJA	102	102.0000
PRD-05D97KHISO	435	435.0000
PRD-05IQWF0GFH	457	457.0000
PRD-05JD815T72	162	162.0000
PRD-05QGSCOH6R	72	72.0000
PRD-065G4PPB1C	22	22.0000
PRD-066RY71JPO	132	132.0000
PRD-06B62PW06Q	445	445.0000
PRD-06MY4JI5X8	327	327.0000
PRD-07QWG4KIMR	343	343.0000
PRD-07Y48UAWVL	106	106.0000
PRD-08HNAX5PM8	131	131.0000
PRD-08RZ3BQ9HX	340	340.0000
PRD-08YL3TZACU	212	212.0000
PRD-092NHUMTMF	247	247.0000
PRD-09H2RXMX2G	178	178.0000

-- 6

SELECT order_id,

LAG(order_value) OVER (ORDER BY order_date)

FROM logistics_erp;

```
mysql> SELECT order_id,  
-> LAG(order_value) OVER (ORDER BY order_date)  
-> FROM logistics_erp;
```

order_id	LAG(order_value) OVER (ORDER BY order_date)
ORD-26-825324	NULL
ORD-26-882775	50587.42
ORD-26-104236	59257.44
ORD-26-563863	148623.36
ORD-26-807731	54989.04
ORD-26-459760	93548.97
ORD-26-724449	11103.22
ORD-26-844152	66362.36
ORD-26-987169	132221.88
ORD-26-288897	7549.50
ORD-26-107633	187489.00
ORD-26-109915	189094.95
ORD-26-603472	186879.54
ORD-26-636608	225396.54
ORD-26-656558	195389.50
ORD-26-479352	114873.60
ORD-26-542349	86189.64
ORD-26-831874	211166.76
ORD-26-844441	65096.91
ORD-26-875520	285958.17
ORD-26-530342	130910.04
ORD-26-447457	16641.99
ORD-26-950184	313255.35
ORD-26-987103	224501.64
ORD-26-712681	24016.27
ORD-26-600208	86336.95
ORD-26-570531	78097.60
ORD-26-834468	77340.78
ORD-26-643798	25743.63
ORD-26-404134	15090.30
ORD-26-865474	86330.25
ORD-26-174731	395529.46

-- 7

SELECT order_id,

LEAD(order_value) OVER (ORDER BY order_date)

FROM logistics_erp;

```
mysql> SELECT order_id,  
-> LEAD(order_value) OVER (ORDER BY order_date)  
-> FROM logistics_erp;
```

order_id	LEAD(order_value) OVER (ORDER BY order_date)
ORD-26-825324	59257.44
ORD-26-882775	148623.36
ORD-26-104236	54989.04
ORD-26-563863	93548.97
ORD-26-807731	11103.22
ORD-26-459760	66362.36
ORD-26-724449	132221.88
ORD-26-844152	7549.50
ORD-26-987169	187489.00
ORD-26-288897	189094.95
ORD-26-107633	186879.54
ORD-26-109915	225396.54
ORD-26-603472	195389.50
ORD-26-636608	114873.60
ORD-26-656558	86189.64
ORD-26-479352	211166.76
ORD-26-542349	65096.91
ORD-26-831874	285958.17
ORD-26-844441	130910.04
ORD-26-875520	16641.99
ORD-26-530342	313255.35
ORD-26-447457	224501.64
ORD-26-950184	24016.27
ORD-26-987103	86336.95
ORD-26-712681	78097.60
ORD-26-600208	77340.78
ORD-26-570531	25743.63
ORD-26-834468	15090.30
ORD-26-643798	86330.25
ORD-26-404134	395529.46
ORD-26-865474	338423.40
ORD-26-174731	2542.70
ORD-26-807111	111344.02

-- 8

```
SELECT customer_id, order_value,  
  
MAX(order_value) OVER (PARTITION BY customer_id)  
  
FROM logistics_erp;
```

```
mysql> SELECT customer_id, order_value,  
-> MAX(order_value) OVER (PARTITION BY customer_id)  
-> FROM logistics_erp;
```

customer_id	order_value	MAX(order_value) OVER (PARTITION BY customer_id)
CUST000066	92052.50	465515.64
CUST000066	11281.40	465515.64
CUST000066	42028.47	465515.64
CUST000066	155524.86	465515.64
CUST000066	145783.29	465515.64
CUST000066	6887.04	465515.64
CUST000066	23516.64	465515.64
CUST000066	12110.72	465515.64
CUST000066	114653.60	465515.64
CUST000066	100981.85	465515.64
CUST000066	137353.09	465515.64
CUST000066	372863.66	465515.64
CUST000066	98638.54	465515.64
CUST000066	309895.44	465515.64
CUST000066	64583.50	465515.64
CUST000066	79474.50	465515.64
CUST000066	78097.60	465515.64
CUST000066	8495.24	465515.64
CUST000066	5010.62	465515.64
CUST000066	43512.66	465515.64
CUST000066	62711.64	465515.64
CUST000066	11782.43	465515.64
CUST000066	3982.00	465515.64
CUST000066	35512.61	465515.64
CUST000066	185988.16	465515.64
CUST000066	171045.54	465515.64
CUST000066	128585.60	465515.64
CUST000066	36847.38	465515.64
CUST000066	32178.30	465515.64
CUST000066	140794.50	465515.64

-- 9

```
SELECT product_id, order_qty,  
  
MIN(order_qty) OVER (PARTITION BY product_id)  
  
FROM logistics_erp;
```

```
mysql> SELECT product_id, order_qty,  
-> MIN(order_qty) OVER (PARTITION BY product_id)  
-> FROM logistics_erp;
```

product_id	order_qty	MIN(order_qty) OVER (PARTITION BY product_id)
PRD-003IZB3MCR	51	51
PRD-013GAF4IU1	136	136
PRD-019SY5F4B8	205	205
PRD-01WEVBW8EN	99	99
PRD-025NU81D1A	325	325
PRD-02Q4J0YQ8R	396	396
PRD-03D65YT8T0	63	63
PRD-03DU22NKPU	447	447
PRD-03KSPFVRF4	479	479
PRD-03LHT5TDTZ	438	438
PRD-03ZA5SNVA4	281	281
PRD-054XDI5YJA	102	102
PRD-05D97KHISO	435	435
PRD-05IQWFOGFH	457	457
PRD-05JD815T72	162	162
PRD-05QGSC0H6R	72	72
PRD-065G4PPB1C	22	22
PRD-066RY71JPO	132	132
PRD-06B62PW06Q	445	445
PRD-06MY4JI5X8	327	327
PRD-07QWG4KIMR	343	343
PRD-07Y48UAWVL	106	106
PRD-08HNAX5PM8	131	131
PRD-08RZ3BQ9HX	340	340
PRD-08YL3TZACU	212	212
PRD-092MHUMTME	217	217

-- 10

SELECT courier,

COUNT(*) OVER (PARTITION BY courier)

FROM logistics_erp;

```
mysql> SELECT courier,  
-> COUNT(*) OVER (PARTITION BY courier)  
-> FROM logistics_erp;
```

[illegible]

-- 11

SELECT warehouse,

SUM(closing_qty) OVER (PARTITION BY warehouse)

FROM logistics_erp;

```
mysql> SELECT warehouse,  
-> SUM(closing_qty) OVER (PARTITION BY warehouse)  
-> FROM logistics_erp;
```

[illegible]

-- 12

SELECT order_date,

SUM(order_value) OVER (ORDER BY order_date)

FROM logistics_erp;

```
mysql> SELECT order_date,
        -> AVG(order_value) OVER (ORDER BY order_date)
        -> FROM logistics_erp;
```

order_date	AVG(order_value) OVER (ORDER BY order_date)
2026-01-01	128379.295417
2026-01-01	128379.295417
2026-01-01	128379.295417
2026-01-01	128379.295417
2026-01-01	128379.295417
2026-01-01	128379.295417
2026-01-01	128379.295417
2026-01-01	128379.295417
2026-01-01	128379.295417
2026-01-01	128379.295417
2026-01-01	128379.295417
2026-01-01	128379.295417
2026-01-01	128379.295417
2026-01-01	128379.295417
2026-01-01	128379.295417
2026-01-01	128379.295417
2026-01-01	128379.295417
2026-01-01	128379.295417
2026-01-01	128379.295417
2026-01-01	128379.295417
2026-01-01	128379.295417
2026-01-01	128379.295417
2026-01-01	128379.295417
2026-01-01	128379.295417
2026-01-01	128379.295417
2026-01-02	121089.310238
2026-01-02	121089.310238
2026-01-02	121089.310238

-- 13

SELECT product_category,

RANK() OVER (PARTITION BY product_category ORDER BY
order_value DESC)

FROM logistics_erp;

```
mysql> SELECT product_category,  
-> RANK() OVER (PARTITION BY product_category ORDER BY order_value DESC)  
-> FROM logistics_erp;
```

product_category	RANK() OVER (PARTITION BY product_category ORDER BY order_value DESC)
Automotive	1
Automotive	2
Automotive	3
Automotive	4
Automotive	5
Automotive	6
Automotive	7
Automotive	8
Automotive	9
Automotive	10
Automotive	11
Automotive	12
Automotive	13
Automotive	14
Automotive	15
Automotive	16
Automotive	17
Automotive	18
Automotive	19
Automotive	20
Automotive	21
Automotive	22
Automotive	23
Automotive	24
Automotive	25

-- 14

SELECT customer_id,

ROW_NUMBER() OVER (PARTITION BY customer_id ORDER BY
order_date)

FROM logistics_erp;

```
mysql> SELECT customer_id,  
-> ROW_NUMBER() OVER (PARTITION BY customer_id ORDER BY order_date)  
-> FROM logistics_erp;
```

customer_id	ROW_NUMBER() OVER (PARTITION BY customer_id ORDER BY order_date)
CUST000066	1
CUST000066	2
CUST000066	3
CUST000066	4
CUST000066	5
CUST000066	6
CUST000066	7
CUST000066	8
CUST000066	9
CUST000066	10
CUST000066	11
CUST000066	12
CUST000066	13
CUST000066	14
CUST000066	15
CUST000066	16
CUST000066	17
CUST000066	18
CUST000066	19
CUST000066	20
CUST000066	21
CUST000066	22
CUST000066	23
CUST000066	24
CUST000066	25
CUST000066	26
CUST000066	27

10 Dynamic SQL Queries

-- 1

```
CREATE PROCEDURE GetCustomerOrders(IN cust_id VARCHAR(20))  
  
BEGIN  
  
    SELECT * FROM logistics_erp WHERE customer_id = cust_id;  
  
END$$
```

-- 2

```
CREATE PROCEDURE GetOrdersByDate(IN start_date DATE, IN  
end_date DATE)  
  
BEGIN  
  
    SELECT * FROM logistics_erp  
  
    WHERE order_date BETWEEN start_date AND end_date;  
  
END$$
```

-- 3

```
CREATE PROCEDURE GetTopCustomers()  
  
BEGIN
```

```
SELECT customer_id, SUM(order_value) total_sales
FROM logistics_erp
GROUP BY customer_id
ORDER BY total_sales DESC
LIMIT 10;
END$$
```

-- 4

```
CREATE PROCEDURE GetProductSales(IN prod_id VARCHAR(20))
BEGIN
    SELECT product_id, SUM(order_qty) total_qty
    FROM logistics_erp
    WHERE product_id = prod_id
    GROUP BY product_id;
END$$
```

-- 5

```
CREATE PROCEDURE GetCourierPerformance()
BEGIN
    SELECT courier,
        AVG(DATEDIFF(delivered_date, dispatch_date)) avg_delivery_days
```

```
FROM logistics_erp  
  
GROUP BY courier;  
  
END$$
```

```
-- 6
```

```
CREATE PROCEDURE GetWarehouseStock()  
  
BEGIN  
  
    SELECT warehouse, SUM(closing_qty) total_stock  
  
    FROM logistics_erp  
  
    GROUP BY warehouse;  
  
END$$
```

```
-- 7
```

```
CREATE PROCEDURE GetDelayedOrders(IN days_limit INT)  
  
BEGIN  
  
    SELECT order_id,  
  
           DATEDIFF(delivered_date, dispatch_date) delay  
  
    FROM logistics_erp  
  
    WHERE DATEDIFF(delivered_date, dispatch_date) > days_limit;  
  
END$$
```

-- 8

CREATE PROCEDURE GetCategorySales()

BEGIN

SELECT product_category, SUM(order_value) total_sales

FROM logistics_erp

GROUP BY product_category;

END\$\$

-- 9

CREATE PROCEDURE UpdateShipmentStatus(

IN ord_id VARCHAR(20),

IN status_val VARCHAR(50)

)

BEGIN

UPDATE logistics_erp

SET shipment_status = status_val

WHERE order_id = ord_id;

END\$\$

-- 10

CREATE PROCEDURE GetMonthlySales()

BEGIN

SELECT MONTH(order_date) month,

SUM(order_value) total_sales

FROM logistics_erp

GROUP BY MONTH(order_date);

END\$\$

DELIMITER ;

10 Stored Procedures

```
DELIMITER $$
```

```
-- 1
```

```
CREATE FUNCTION GetDeliveryDays(d DATE, r DATE)
```

```
RETURNS INT
```

```
DETERMINISTIC
```

```
BEGIN
```

```
    RETURN DATEDIFF(r, d);
```

```
END$$
```

```
-- 2
```

```
CREATE FUNCTION GetDiscount(val DECIMAL(10,2))
```

```
RETURNS DECIMAL(10,2)
```

```
DETERMINISTIC
```

```
BEGIN
```



```
    RETURN val * 0.9;
```

```
END$$
```

```
-- 3
```

```
CREATE FUNCTION GetTax(val DECIMAL(10,2))
```

```
RETURNS DECIMAL(10,2)
```

```
DETERMINISTIC
```

```
BEGIN
```

```
    RETURN val * 0.18;
```

```
END$$
```

```
-- 4
```

```
CREATE FUNCTION NetAmount(val DECIMAL(10,2))
```

```
RETURNS DECIMAL(10,2)
```

```
DETERMINISTIC
```

```
BEGIN
```

```
    RETURN val - (val * 0.1);
```

```
END$$
```

```
-- 5
```

```
CREATE FUNCTION IsDelayed(d DATE, r DATE)
```

```
RETURNS VARCHAR(10)

DETERMINISTIC

BEGIN

    IF DATEDIFF(r, d) > 5 THEN

        RETURN 'YES';

    ELSE

        RETURN 'NO';

    END IF;

END$$
```

```
-- 6
```

```
CREATE FUNCTION CategoryLabel(cat VARCHAR(50))

RETURNS VARCHAR(50)

DETERMINISTIC

BEGIN

    RETURN CONCAT('Category: ', cat);

END$$
```

```
-- 7
```

```
CREATE FUNCTION DoubleQty(q INT)

RETURNS INT
```

DETERMINISTIC

BEGIN

RETURN q * 2;

END\$\$

-- 8

CREATE FUNCTION AvgPrice(val1 DECIMAL(10,2), val2
DECIMAL(10,2))

RETURNS DECIMAL(10,2)

DETERMINISTIC

BEGIN

RETURN (val1 + val2) / 2;

END\$\$

-- 9

CREATE FUNCTION YearFromDate(d DATE)

RETURNS INT

DETERMINISTIC

BEGIN

RETURN YEAR(d);

END\$\$

-- 10

CREATE FUNCTION OrderValueCategory(val DECIMAL(10,2))

RETURNS VARCHAR(20)

DETERMINISTIC

BEGIN

IF val > 10000 THEN

RETURN 'HIGH';

ELSEIF val > 5000 THEN

RETURN 'MEDIUM';

ELSE

RETURN 'LOW';

END IF;

END\$\$

DELIMITER ;

- 10 User Defined Functions

10 User Defined Functions

-- 1

CREATE FUNCTION GetDeliveryDays(d DATE, r DATE)

RETURNS INT

DETERMINISTIC

BEGIN

 RETURN DATEDIFF(r, d);

END\$\$

-- 2

CREATE FUNCTION GetDiscount(val DECIMAL(10,2))

RETURNS DECIMAL(10,2)

DETERMINISTIC

BEGIN

 RETURN val * 0.9;

END\$\$

-- 3

CREATE FUNCTION GetTax(val DECIMAL(10,2))

RETURNS DECIMAL(10,2)

DETERMINISTIC

BEGIN

 RETURN val * 0.18;

END\$\$

-- 4

CREATE FUNCTION NetAmount(val DECIMAL(10,2))

RETURNS DECIMAL(10,2)

DETERMINISTIC

BEGIN

 RETURN val - (val * 0.1);

END\$\$

-- 5

CREATE FUNCTION IsDelayed(d DATE, r DATE)

RETURNS VARCHAR(10)

DETERMINISTIC

BEGIN

```
IF DATEDIFF(r, d) > 5 THEN  
    RETURN 'YES';  
  
ELSE  
    RETURN 'NO';  
  
END IF;  
END$$
```

```
-- 6
```

```
CREATE FUNCTION CategoryLabel(cat VARCHAR(50))  
RETURNS VARCHAR(50)  
DETERMINISTIC  
BEGIN  
    RETURN CONCAT('Category: ', cat);  
END$$
```

```
-- 7
```

```
CREATE FUNCTION DoubleQty(q INT)  
RETURNS INT  
DETERMINISTIC  
BEGIN  
    RETURN q * 2;
```

```
END$$
```

```
-- 8
```

```
CREATE FUNCTION AvgPrice(val1 DECIMAL(10,2), val2  
DECIMAL(10,2))
```

```
RETURNS DECIMAL(10,2)
```

```
DETERMINISTIC
```

```
BEGIN
```

```
    RETURN (val1 + val2) / 2;
```

```
END$$
```

```
-- 9
```

```
CREATE FUNCTION YearFromDate(d DATE)
```

```
RETURNS INT
```

```
DETERMINISTIC
```

```
BEGIN
```

```
    RETURN YEAR(d);
```

```
END$$
```

```
-- 10
```

```
CREATE FUNCTION OrderValueCategory(val DECIMAL(10,2))
```

```
RETURNS VARCHAR(20)
```


DETERMINISTIC

BEGIN

IF val > 10000 THEN

 RETURN 'HIGH';

ELSEIF val > 5000 THEN

 RETURN 'MEDIUM';

ELSE

 RETURN 'LOW';

END IF;

END\$\$

DELIMITER ;

10 Views

-- 1

```
CREATE VIEW view_customer_sales AS  
SELECT customer_id, SUM(order_value) total_sales  
FROM logistics_erp  
GROUP BY customer_id;
```

-- 2

```
CREATE VIEW view_product_sales AS  
SELECT product_id, SUM(order_qty) total_qty  
FROM logistics_erp  
GROUP BY product_id;
```

-- 3

```
CREATE VIEW view_courier_performance AS  
SELECT courier,  
       AVG(DATEDIFF(delivered_date, dispatch_date)) avg_delivery  
FROM logistics_erp  
GROUP BY courier;
```

-- 4

```
CREATE VIEW view_category_sales AS  
  
SELECT product_category, SUM(order_value) total_sales  
  
FROM logistics_erp  
  
GROUP BY product_category;
```

-- 5

```
CREATE VIEW view_warehouse_stock AS  
  
SELECT warehouse, SUM(closing_qty) total_stock  
  
FROM logistics_erp  
  
GROUP BY warehouse;
```

-- 6

```
CREATE VIEW view_delayed_orders AS  
  
SELECT order_id,  
  
       DATEDIFF(delivered_date, dispatch_date) delay  
  
FROM logistics_erp  
  
WHERE DATEDIFF(delivered_date, dispatch_date) > 5;
```

-- 7

```
CREATE VIEW view_monthly_sales AS
```

```
SELECT MONTH(order_date) month,  
       SUM(order_value) total_sales  
FROM logistics_erp  
GROUP BY MONTH(order_date);
```

-- 8

```
CREATE VIEW view_top_orders AS  
SELECT order_id, order_value  
FROM logistics_erp  
ORDER BY order_value DESC;
```

-- 9

```
CREATE VIEW view_customer_orders_count AS  
SELECT customer_id, COUNT(order_id) total_orders  
FROM logistics_erp  
GROUP BY customer_id;
```

-- 10

```
CREATE VIEW view_product_avg_price AS  
SELECT product_id, AVG(order_value) avg_price  
FROM logistics_erp  
GROUP BY product_id;
```

5 Materialized Views

-- 1

```
CREATE TABLE mv_customer_sales AS  
SELECT customer_id, SUM(order_value) total_sales  
FROM logistics_erp GROUP BY customer_id;
```

-- 2

```
CREATE TABLE mv_product_sales AS  
SELECT product_id, SUM(order_qty) total_qty  
FROM logistics_erp GROUP BY product_id;
```

-- 3

```
CREATE TABLE mv_courier_stats AS  
SELECT courier, COUNT(*) total_orders  
FROM logistics_erp GROUP BY courier;
```

-- 4

```
CREATE TABLE mv_warehouse_stock AS
```

```
SELECT warehouse, SUM(closing_qty) total_stock  
FROM logistics_erp GROUP BY warehouse;
```

```
-- 5
```

```
CREATE TABLE mv_daily_sales AS  
SELECT order_date, SUM(order_value) sales  
FROM logistics_erp GROUP BY order_date;
```

5 MERGE Statements

```
-- 1
```

```
INSERT INTO summary_table (customer_id, total_sales)  
SELECT customer_id, SUM(order_value)  
FROM logistics_erp GROUP BY customer_id  
ON DUPLICATE KEY UPDATE total_sales = VALUES(total_sales);
```

```
-- 2
```

```
INSERT INTO product_summary (product_id, total_qty)  
SELECT product_id, SUM(order_qty)  
FROM logistics_erp GROUP BY product_id
```

```
ON DUPLICATE KEY UPDATE total_qty = VALUES(total_qty);
```

```
-- 3
```

```
INSERT INTO courier_summary (courier, total_orders)
```

```
SELECT courier, COUNT(*)
```

```
FROM logistics_erp GROUP BY courier
```

```
ON DUPLICATE KEY UPDATE total_orders = VALUES(total_orders);
```

```
-- 4
```

```
INSERT INTO warehouse_summary (warehouse, total_stock)
```

```
SELECT warehouse, SUM(closing_qty)
```

```
FROM logistics_erp GROUP BY warehouse
```

```
ON DUPLICATE KEY UPDATE total_stock = VALUES(total_stock);
```

```
-- 5
```

```
INSERT INTO daily_summary (order_date, sales)
```

```
SELECT order_date, SUM(order_value)
```

```
FROM logistics_erp GROUP BY order_date
```

```
ON DUPLICATE KEY UPDATE sales = VALUES(sales);
```

5 Recursive CTEs

-- 1

```
WITH RECURSIVE seq AS (  
    SELECT 1 n  
    UNION ALL  
    SELECT n+1 FROM seq WHERE n < 10  
)  
SELECT * FROM seq;
```

-- 2

```
WITH RECURSIVE dates AS (  
    SELECT MIN(order_date) d FROM logistics_erp  
    UNION ALL  
    SELECT DATE_ADD(d, INTERVAL 1 DAY)  
    FROM dates WHERE d < (SELECT MAX(order_date) FROM  
    logistics_erp)  
)  
SELECT * FROM dates;
```

-- 3


```
WITH RECURSIVE nums AS (  
    SELECT 1 n  
    UNION ALL  
    SELECT n+2 FROM nums WHERE n < 20  
)  
SELECT * FROM nums;
```

-- 4

```
WITH RECURSIVE factorial AS (  
    SELECT 1 n, 1 fact  
    UNION ALL  
    SELECT n+1, fact*(n+1) FROM factorial WHERE n < 5  
)  
SELECT * FROM factorial;
```

-- 5

```
WITH RECURSIVE hierarchy AS (  
    SELECT customer_id, 1 lvl FROM logistics_erp  
    UNION ALL  
    SELECT customer_id, lvl+1 FROM hierarchy WHERE lvl < 3  
)
```

```
SELECT * FROM hierarchy;
```